

Deferred Related Work candidate. This is not the full manuscript.

1 Related Work

Dense dual encoders rank a corpus through vector similarity. BEIR tests transfer across retrieval tasks (Thakur et al., 2021); DenseOn supplies the pretrained checkpoint and open adaptation recipe (Sourty et al., 2026). HIL combines isotropic and anisotropic representations in late-interaction retrieval (Kim et al., 2024), motivating relevance-sensitive evaluation rather than maximizing a spread statistic alone.

Takeshita et al. (2025) study random and single-coordinate removal in text embeddings. Wu et al. (2026) learn query-specific dimension selection from positive-negative alignment. Relevance-sensitive dimension importance is therefore not itself our contribution. We track finite deletion sensitivities along optimizer trajectories and test whether their native-coordinate differences survive shared rotations and predict full-corpus retrieval.

Muon scales well in language-model pretraining, and NorMuon adds neuron-wise adaptation (Liu et al., 2025; Li et al., 2025). Associative-memory and imbalance studies connect spectral updates to more balanced learning (Wang et al., 2025; Vasudeva et al., 2025); feature-transfer benefits have also been reported (Ruan et al., 2026). These are functional explanations, not just descriptions of an update rule. Yet spectral descent can lose useful simplicity bias (Dragutinović and Ranganath, 2026), and optimizer switching can disrupt pretrained knowledge (Qu et al., 2026; Liu et al., 2026). Spectral correction, input covariance, and layer-dependent geometry offer further qualifications (Pang et al., 2026; Du and Su, 2026; Massena et al., 2026).

Basis and scale require separate care. Outlier-safe pretraining combines Muon with architectural changes (Park et al., 2025); parameter-gauge equivariance is distinct from our post-encoding rotations (Singh, 2026). Hyperball connects weight and update norms to angular dynamics (Wen et al., 2026). Our initial fixed-probe calibration does not match entire trajectories. We ask which reached-state measurements predict held-out-rate retrieval beyond the training recipe, then separately cross fixed source states with reset continuation rules. Neither a spectral fingerprint nor a basis-sensitive coordinate effect alone establishes a retrieval mechanism.

References

- Sara Dragutinović and Rajesh Ranganath. 2026. [To use or not to use Muon: How simplicity bias in optimizers matters.](#) *arXiv preprint arXiv:2603.00742*.
- Zhehang Du and Weijie Su. 2026. [The Newton–Muon optimizer.](#) *arXiv preprint arXiv:2604.01472*.
- Jaeyoung Kim, Dohyeon Lee, and Seung-won Hwang. 2024. [HIL: Hybrid isotropy learning for zero-shot performance in dense retrieval.](#) In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 7892–7903. Association for Computational Linguistics.
- Zichong Li, Liming Liu, Chen Liang, Weizhu Chen, and Tuo Zhao. 2025. [NorMuon: Making Muon more efficient and scalable.](#) *arXiv preprint arXiv:2510.05491*.
- Jingyuan Liu, Jianlin Su, Xingcheng Yao, Zhejun Jiang, Guokun Lai, Yulun Du, Yidao Qin, Weixin Xu, Enzhe Lu, Junjie Yan, et al. 2025. [Muon is scalable for LLM training.](#) *arXiv preprint arXiv:2502.16982*.
- Yuxing Liu, Jianyu Wang, and Tong Zhang. 2026. [Optimizer-model consistency: Full finetuning with the same optimizer as pretraining forgets less.](#) *arXiv preprint arXiv:2605.06654*.
- Thomas Massena, Corentin Friedrich, and Mathieu Serurier. 2026. [From SGD to Muon: Adaptive optimization via Schatten-p norms.](#) *arXiv preprint arXiv:2605.19781*.
- Tianyu Pang, Yujie Fang, Zihang Liu, Shenyang Deng, Lei Hsiung, Shuhua Yu, and Yaoqing Yang. 2026. [HTMuon: Improving Muon via heavy-tailed spectral correction.](#) In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 36504–36535. Association for Computational Linguistics.
- Jungwoo Park, Taewhoo Lee, Chanwoong Yoon, Hyeon Hwang, and Jaewoo Kang. 2025. [Outlier-safe pretraining for robust 4-bit quantization of large language models.](#) In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 12582–12600. Association for Computational Linguistics.
- Xingyu Qu, Peigeng Huang, and Samuel Horvath. 2026. [Can Muon fine-tune Adam-pretrained models?](#) *arXiv preprint arXiv:2605.10468*.
- Tianyu Ruan, Fengzhuo Zhang, Shuche Wang, and Shihua Zhang. 2026. [Muon learns more robust and transferable features than Adam.](#) *arXiv preprint arXiv:2606.09658*.
- Devender Singh. 2026. [The loss does not see the basis, but Adam does.](#) *arXiv preprint arXiv:2608.05136*.
- Raphaël Sourty, Antoine Chaffin, Paulo Roberto Moura Junior, and Amélie Chatelain. 2026. [DenseOn](#)

with the LateOn: Fully open dense and late-interaction models for multilingual, long-context, and code search. *arXiv preprint arXiv:2607.27178*.

Sotaro Takeshita, Yurina Takeshita, Daniel Ruffinelli, and Simone Paolo Ponzetto. 2025. Randomly removing 50% of dimensions in text embeddings has minimal impact on retrieval and classification tasks. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 27705–27726. Association for Computational Linguistics.

Nandan Thakur, Nils Reimers, Andreas Rücklé, Abhishek Srivastava, and Iryna Gurevych. 2021. BEIR: A heterogeneous benchmark for zero-shot evaluation of information retrieval models. In *Thirty-fifth Conference on Neural Information Processing Systems Datasets and Benchmarks Track*.

Bhavya Vasudeva, Puneesh Deora, Yize Zhao, Vatsal Sharan, and Christos Thrampoulidis. 2025. How Muon’s spectral design benefits generalization: A study on imbalanced data. *arXiv preprint arXiv:2510.22980*.

Shuche Wang, Fengzhuo Zhang, Jiaxiang Li, Cunxiao Du, Chao Du, Tianyu Pang, Zhuoran Yang, Mingyi Hong, and Vincent Y. F. Tan. 2025. Muon outperforms Adam in tail-end associative memory learning. *arXiv preprint arXiv:2509.26030*.

Kaiyue Wen, Xingyu Dang, Kaifeng Lyu, Tengyu Ma, and Percy Liang. 2026. Fantastic pretraining optimizers and where to find them II: Hyperball optimization. *arXiv preprint arXiv:2606.16899*.

Zhanyu Wu, Richong Zhang, and Zhijie Nie. 2026. Learning to select: Query-aware adaptive dimension selection for dense retrieval. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 18666–18677. Association for Computational Linguistics.

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